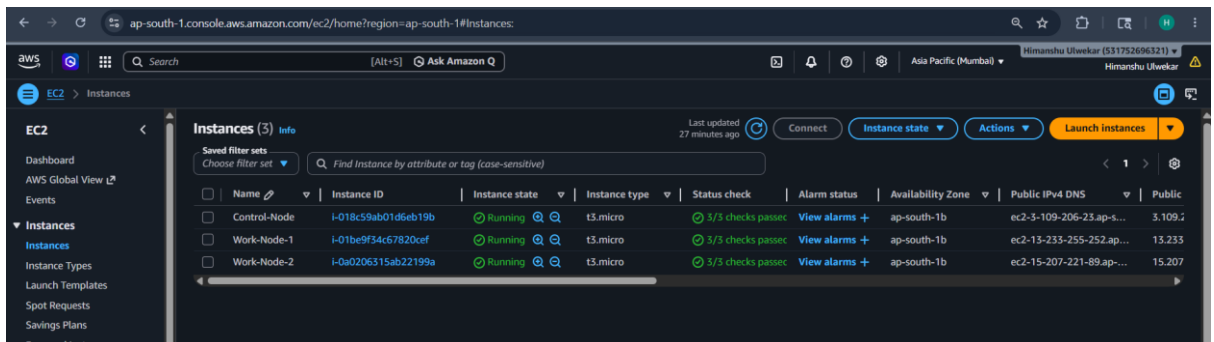


✓ STEP 1 — EC2 Setup (Already Done in Your Case)

You should have:

- 1 Control Node (Ansible installed here)
- 2 Worker Nodes
- Same Key Pair used for all instances
- Security Group allows SSH (port 22)



✓ STEP 2 — Install Ansible (Control Node)

```
sudo yum update -y
sudo yum install ansible -y
```

Check:

```
ansible --version
```

✓ STEP 3 — Install AWS dependencies

```
sudo yum install python3-pip
pip3 install boto3 botocore
ansible-galaxy collection install amazon.aws
```

✓ STEP 4 — Configure AWS credentials

```
aws configure
```

✓ **STEP 5— Fix PEM file on control node**

Copy key (if not already there):

It is better to copy pem file in ansible directory.

On control node:

```
sudo chown ec2-user:ec2-user /etc/ansible/dynamic.pem
```

```
sudo chmod 400 /etc/ansible/dynamic.pem
```

✓ **STEP 7 — Go to the ansible directory**

```
cd ansible
```

✓ **STEP 8 — Create AWS Dynamic Inventory file**

```
Sudo vi aws_ec2.yml
```

```
plugin: amazon.aws.aws_ec2
```

```
regions:  
  - ap-south-1
```

```
hostnames:  
  - private-ip-address
```

```
filters:  
  instance-state-name: running
```

```
compose:  
  ansible_host: private_ip_address  
  ansible_user: ec2-user  
  ansible_ssh_private_key_file: /etc/ansible/dynamic.pem
```

```
keyed_groups:  
  - key: tags.Name  
    prefix: ""
```

✓ **STEP 9 — Create ansible.cfg**

```
sudo vi ansible.cfg
```

Paste:

```
[defaults]
inventory = aws_ec2.yml
private_key_file = /etc/ansible/dynamic.pem
host_key_checking = False
[inventory]
enable_plugins = amazon.aws.aws_ec2
```



STEP 10 — Test inventory discovery

```
ansible-inventory --graph
```

Expected:

```
_Work_Node_1
_Work_Node_2
_Control_Node
```



STEP 11 — Test SSH connectivity

Single node:

```
ansible -i aws_ec2.yml _Work_Node_1 -m ping
```

OR all workers:

```
ansible -i aws_ec2.yml _Work_Node_* -m ping
```



FINAL EXPECTED OUTPUT

```
SUCCESS => {
  "ping": "pong"
}
```



WHAT YOU HAVE NOW

- ✓ Fully working AWS dynamic inventory
- ✓ Auto-discovered EC2 instances
- ✓ Tag-based grouping

- ✓ SSH working with PEM
 - ✓ Ansible remote execution ready
-

IMPORTANT RULES (don't break it again)

- ✗ Don't use `root` user
- ✗ Don't hardcode `workers` group unless you define it properly
- ✗ Don't rely on inventory for key file alone
- ✓ Always ensure `ec2-user` + correct `.pem`